

Задание 19 - Кросс-компиляция

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1) Собраем конфигурацию под x86_64

```
svikari@svikari-VirtualBox:~/zlib-1.3.2$ ./configure -prefix=$PWD/_instal -static
Checking for gcc...
Checking for shared library support...
Building shared library libz.so.1.3.2 with gcc.
Checking for size_t... Yes.
Checking for off64_t... Yes.
Checking for fseeko... Yes.
Checking for strerror... Yes.
Checking for unistd.h... Yes.
Checking for stdarg.h... Yes.
Checking whether to use vs[n]printf() or s[n]printf()... using vs[n]printf().
Checking for vsnprintf() in stdio.h... Yes.
Checking for return value of vsnprintf()... Yes.
Checking for attribute(visibility) support... Yes.
Checking for s390x build ... No.
```

2) Собраем конфигурацию под arm

```
svikari@svikari-VirtualBox:~/zlib-1.3.2$ make CC=arm-linux-gnueabi-gcc
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o example.o test/example.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o adler32.o adler32.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o crc32.o crc32.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o deflate.o deflate.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o infback.o infback.c
```

```
svikari@svikari-VirtualBox:~/zlib-1.3.2$ file trees.o
trees.o: ELF 32-bit LSB relocatable, ARM, EABI5 version 1 (SYSV), not stripped
```

3) Запускаем конфигурацию с zlib

```
svikari@svikari-VirtualBox:~/zlib-1.3.2$ make CC=arm-linux-gnueabi-gcc
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o example.o test/example.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o adler32.o adler32.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o crc32.o crc32.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o deflate.o deflate.c
arm-linux-gnueabi-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o infback.o infback.c
```

```
svikari@svikari-VirtualBox:~/openssl-9.7p1$ ./configure -host=arm-linux-gnueabi
hf -with-zlib=$PWD/../../zlib-1.3.2/_install -prefix=$PWD/_install
checking for arm-linux-gnueabi-hf-cc... no
checking for arm-linux-gnueabi-hf-gcc... arm-linux-gnueabi-hf-gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... yes
checking for suffix of object files... o
checking whether the compiler supports GNU C... yes
checking whether arm-linux-gnueabi-hf-gcc accepts -g... yes
checking for arm-linux-gnueabi-hf-gcc option to enable C11 features... none needed
checking if arm-linux-gnueabi-hf-gcc supports C99-style variadic macros... yes
checking build system type... x86_64-pc-linux-gnu
```

4) Устанавливаем SSL

```
svikari@svikari-VirtualBox:~$ wget https://www.openssl.org/source/openssl-3.0.13
.tar.gz
tar -xzf openssl-3.0.13.tar.gz
cd openssl-3.0.13
```

5) Запускаем QEMU

```
svikari@svikari-VirtualBox:~/lab19$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpr
ess-a9 -kernel zImage -initrd initramfs_ssh.cpio.gz -dtb vexpress-v2p-c9.dtb -ap
pend "console=ttyAMA0 rdinit=/bin/ash" -nographic
```

```
~ #
~ # ls
bin          initramfs.cpio.gz  sbin
dev          linuxrc            usr
etc          root               var
```